

Mohamed Riyasudeen D

Embedded Software Developer

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Professional Experience

Embedded Software Developer (BMS),

IElektron Technologies (On-site at Royal Enfield)

09/2023 – Present
Chennai

- Controller - NXP S32K322.
- Developed production CDD drivers for ADBMS6832 (cell monitor) & ADBM2950 (pack monitor) with functional safety; implemented cell voltage, NTC temperature, shunt current, open-wire detection, and IMD isolation calculations.
- Implemented passive cell balancing logic (voltage thresholds, PWM control) in charge and sleep modes; reduced cell delta voltage by 10-20 mV during charge cycles & below 10 mV in sleep mode.
- Worked on power management in BMS using PMIC (FS26) and used LPCM of AFEs to monitor cell voltages & temperature in low power mode.
- Built Vector CANoe panels + CAPL automation for BMS parameter monitoring (single-view dashboard) and fault injection; accelerated validation by 40%.
- Supported AIS-156 homologation and EMI/EMC/ESD testing for EV BMS at ARAI Pune and TUV Bangalore, with hands-on understanding of test cases, procedures, and terminology.

Embedded Software Developer (Bootloader/UDS), IElektron Technologies

02/2023 – 09/2023
Chennai

- Controller - Renesas RL78/F13.
- Worked on the EEPROM emulation library of RENESAS & developed the complete NVM manager for data flash & code flash.
- Implemented UDS services (0x10 DiagnosticSessionControl, 0x27 SecurityAccess, 0x22 RDBI, 0x2E WDBI, 0x34/36/37 Download/Transfer) on Renesas RL78F13.

Education

Electrical & Electronics Engineering, Syed Ammal Engineering College

2016 – 2020

Skills

MCU Platforms

NXP S32K322 (S32K3), Renesas RL78/F13, STM32

Quality

MISRA C, Functional Safety, AIS-156 (EV), EMI/EMC

Debugging/Testing

Oscilloscope, Function generator, RPS, Electronic loads, bi-directional supplies, board bring-up, Schematic analysis/ review

Development Tools

S32 Design Studio, Renesas e2 Studio, IAR IDE, Vector CANoe, Git (GitHub/Bitbucket/Git Bash), QT Creator, Visual Studio (DLL development)

Comms/Protocols

CAN, SPI, I2C, UART, UDS (ISO 14229)

Programming

C/Embedded C, FreeRTOS, CAPL scripting, CAPL DLL, C++

Courses

Embedded Systems, *Vector India*

2022 – 2023
Chennai